

Exploring Opportunities to Augment Psychotherapy with Language Models

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ABSTRACT

Natural language data, like patient narratives, are crucial in psychotherapy, yet psychotherapists face challenges using these qualitative data to tailor treatment to patient needs. Innovations in natural language processing, including breakthroughs in language models (LMs), show opportunities like summarizing conversational data into quantitative information. In this study, we investigated how LM-based tools can augment patient measurements and treatment delivery in psychotherapy. Through formative interviews and design provocation sessions with a total of six psychotherapists, we identified three opportunities: 1) to quantify and summarize extensive qualitative data for easier retrieval and monitoring of treatment progress; 2) to give clinicians a structured tool to support their patients' learning; and 3) to facilitate treatment personalization. Our findings suggest that LM-based tools can potentially facilitate data-driven clinical practice, reduce cognitive and administrative burdens, and improve treatment quality. Additionally, our research paves the way for developing LM-based tools that can be integrated into psychotherapy.

CCS CONCEPTS

• **General and reference** → **Surveys and overviews**; • **Human-centered computing** → **HCI design and evaluation methods**; **Interaction design process and methods**; *Empirical studies in interaction design*; *Systems and tools for interaction design*.

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1 INTRODUCTION

Psychotherapy is a collaborative mental health treatment process that helps individuals with mental health conditions understand and resolve psychological and behavioral challenges. While psychotherapy may be administered with different treatment approaches and targets, the primary aims of treatment are to alleviate patients' symptoms and enhance their quality of life [34, 46]. However, common challenges hampering the effectiveness of psychotherapy include accurately measuring complex psychological processes and treatment outcomes [10, 55, 61], ensuring patients' treatment engagement [4, 17, 54], and tailoring treatment to patients' needs [25, 63]. Recent natural language processing (NLP) technologies, including language models (LMs), can process patient narratives for diverse clinical purposes [40] and flexibly interact with natural language data [23]. These technologies could be instrumental in translating content from therapy sessions (e.g., patient-specific treatment goals) [16, 26] into measurable outcomes for more personalized treatment [36], improving treatment efficiency, and hence increasing access to care of higher quality.

Given this motivation, we explored how LMs can assist psychotherapists through a two-part qualitative study with six mental health clinicians from various backgrounds and areas of clinical experience. First, we conducted formative interviews to learn about clinicians' current practice, which informed our following design provocation study to further investigate opportunities for LM-based tools to address identified challenges and augment psychotherapy in practice. We provided participants with an interactive Figma prototype that displayed generated data by GPT-4¹. Participants were asked to express their opinions about how the generated data might be useful in psychotherapy. We identified three opportunities for LMs to augment psychotherapists' practice: (1) to quantify and summarize extensive qualitative data for more precise assessments and monitoring of symptoms, goals, and outcomes; (2) to provide clinicians with a structured tool to support patients' learning; and (3) to facilitate treatment personalization for more targeted interventions. These opportunities create exciting research directions to develop LM-based tools that maximize psychotherapists' use of qualitative patient data and improve the quality of psychotherapy.

¹<https://openai.com/research/gpt-4>

2 BACKGROUND AND RELATED WORK

2.1 Language Processing and Applications in Mental Health

In mental healthcare, natural language exchanges play a central role during and outside psychotherapy sessions [42]. Innovations in Natural Language Processing (NLP), have made it possible to quantitatively study a large amount of free-text or semi-structured natural language data from patient narratives in psychotherapy [33, 37]. Previous research demonstrated effective NLP applications in mental health that identify topics in psychotherapy conversations [47], analyze clinical notes to identify patient emotions within psychotherapy [56], monitor patients' symptoms [38], and map linguistic data to indicators of treatment progress [40], extending the feasibility of leveraging qualitative narratives toward data-driven mental health treatment approaches [11]. More recently, language models have demonstrated a remarkable ability to process and generate natural language [9, 14, 43], creating new opportunities to process conversational data within a psychotherapeutic context. There is extensive research focused on establishing frameworks to fine-tune and evaluate pre-trained language models to improve their domain-specific performance in mental-health-specific tasks such as stress, depression, and suicide detection [32, 62]. Yet, despite a significant technical promise, LM-based tools remain far from real-world translation [62]. In this work, we explored how existing language models can process patient narrative data to improve psychotherapy outcomes.

Language models' flexibility in completing diverse natural language tasks has already inspired research prototyping assistive tools to support mental well-being and psychotherapy treatment progress [27, 48, 51]. Previous studies have demonstrated that a language-model-based AI-in-the-loop systems can enable more empathetic conversations in peer-to-peer mental health support [49]. In another example, MindfulDiary [30] introduces an LM-based chatbot that could assist clinicians to better understand patients' thoughts and daily contexts, though researchers only received feedback from patients, not clinicians. From a design perspective, there is a lack of user-centered studies that engage clinicians to understand how AI-generated insights can assist with day-to-day challenges faced by providers within psychotherapy [6, 31, 57]. In this study, we explored with clinicians the design space of LM-based tools that can process patient narrative data in psychotherapy to collaboratively identify challenges and opportunities. We aimed to generate a more nuanced understanding of how text-based AI applications can be tailored to effectively complement and augment current psychotherapeutic practice.

2.2 Psychotherapy

Psychotherapy, a psychosocial treatment across different clinical settings (e.g., hospital, outpatient clinics, private practices), is administered by *psychotherapists* who are trained in and employ specific therapeutic frameworks, including cognitive-behavioral therapy (CBT), dialectical behavioral therapy (DBT), and psychodynamic therapy [21, 34]. These frameworks use different methodologies to improve psychological outcomes. CBT helps patients understand

and change *maladaptive thoughts and behaviors* that lead to negative emotional states [34]; psychodynamic therapy helps patients *explore and understand unconscious processes and past interpersonal dynamics* that might influence their current behaviors [50]; and DBT, adapted from CBT, helps patients *develop emotion regulation, mindfulness, distress management, and interpersonal skills* [13]. While CBT and DBT follow standardized protocols targeting specific symptoms of mental health disorders [34, 41], psychodynamic therapy is less structured, allowing patients the flexibility to explore unconscious processes [50].

A growing line of work suggests that tailoring *assessment* [19, 20] and *intervention delivery* [21, 44], by balancing standardized treatments with personalized care, might improve treatment relevance and acceptability to patients [5] as well as patient engagement [39] and outcomes [25, 28]. Therapist flexibility in adapting empirically grounded treatments to meet specific patient needs (i.e., *adherence flexibility* [29]) is crucial to effective personalized psychotherapy [5, 28]. However, there is scarce evidence on how to effectively deliver treatments that are both personalized to patient needs and based on evidence-based practices in real time [5, 18]. This highlights the urgent need for novel solutions to help clinicians personalize treatment and improve outcomes.

Furthermore, another barrier to personalizing psychotherapy is the lack of *idiographic* measurements that capture patient-specific symptoms and outcomes. Existing standardized measures (e.g., standardized questionnaires, like the brief Generalized Anxiety Disorder scale (GAD-7 [53]), which focus on general symptoms, fail to capture patients' unique symptom profiles and outcomes, hindering the process of assessing and setting individually-defined treatment goals [26, 35]. In contrast, *idiographic* assessments (e.g., focused situational measures on patients' anxiety levels, for example, in large gatherings) can provide clinicians with tailored, actionable patient-relevant information [52, 58]. Studies show that combining subjective, patient-specific information and standardized measures can help clinicians better evaluate treatment progress in psychotherapy [35, 60]. Given these challenges, we investigated with clinicians future research opportunities and innovative interaction modalities of LM-based tools to support psychotherapists' in-situ needs for delivering more patient-centered care.

3 METHODS

In this qualitative study, we investigated opportunities for LM-based tools to enhance measurement and treatment delivery in psychotherapy. We first conducted formative interviews to understand mental health clinicians' current practice and challenges in assessing patients' symptoms and treatment outcomes. We then used findings from these formative interviews to inform a design provocation session that explored clinicians' perceptions of using LM-based clinical support tools in practice and helped elicit future opportunities to apply language models under the psychotherapy context.

3.1 Study Procedures and Analysis

We conducted formative interviews and design provocation with six mental health clinicians. Five of these participants (all except C33) participated in both sessions. Participants were recruited from

Table 1: Background information on interviewed participants. All participants took part in the formative interviews and design provocation sessions, except C33 only participated in the formative interviews.

ID	Age	Treatment Practices	Clinical (years)	Experience	Patients' (years old)	Age Groups
C33	31	Psychodynamic	3		18-60	
C35	32	CBT, DBT	8		<18; 18-60	
C38	not disclosed	CBT	15		<18; 18-60	
C39	28	Psychodynamic	2		<18; 18-60; >60	
C43	34	CBT, DBT	10		18-60	
C34	35	CBT, ACT	9		<18	

universities in the northeast and southeast United States using snowball sampling [22]. All study procedures were approved by the co-authors' Institutional Review Board (IRB). Participants provided informed consent prior to participation and received a \$30 gift card per one hour session completed. The interviews were recorded with participants' permission, and transcripts from these interviews were de-identified.

In the formative interviews, participants completed a 60-minute interview via Zoom and were asked about their current practice and challenges. Following the formative interviews, the design provocation was conducted during a 30-min Zoom meeting. In this session, we employed interpretive inquiry, which has been used in the HCI community as an effective way to stimulate feedback and discussion [1, 7]. We conducted two activities in the design provocation session to further investigate (1) how clinicians collect and interpret patient narrative data for psychotherapy and (2) clinicians' perception and vision of using LMs to support processing and utilizing patient narratives for measuring patient outcomes and personalizing therapy.

In Activity 1, we aimed to learn about clinicians' thought process when receiving patient narratives. We provided clinicians with a snippet of simulated, transcribed patient utterances on a shared online document. Clinicians were instructed to demonstrate (e.g., highlight, take notes, comment) how they would interpret patient narrative data and to describe their thought process out loud. In Activity 2, we provided participants with an interactive Figma prototype, which simulates an LM-based tool for processing the same patient narrative presented in Activity 1 (see Fig. 1). During the session, participants were asked to explore and interact with the features while thinking out loud. We aimed to create an environment for mental health clinicians to openly react to and provide their insights on the possible future of using LM-based tools to support their clinical practice. Transcripts from the interviews were analyzed using inductive thematic analysis [8]. The first two co-authors independently analyzed and generated open codes for transcripts from the audio recording of the interviews. We then iteratively and collaboratively consolidated the initial open codes into 41 codes and organized them into clusters, creating a description or lower-level theme for each cluster. In the last round

of analysis, we developed summaries or higher-level themes from the previous cluster descriptions.

4 FINDINGS

From our analysis, we uncovered three key opportunities (themes) for LM-based tools to support mental health clinicians delivering psychotherapy: 1) to quantify and summarize extensive qualitative data for easier retrieval and monitoring of treatment progress; 2) to give clinicians a structured tool to support their patients' learning; and 3) to facilitate treatment personalization.

4.1 Opportunity 1: To quantify and summarize qualitative data for monitoring of treatment progress

"Every person's thought spiral is going to be different." (C35) Our participants described how patients' subjective experiences inform clinicians' approach to treatment and help clinicians contextualize results from quantitative measures:

"If I asked you to rate your sadness on a 0-to-10 scale, you might rate yourself as a 4, but you know your mom only beat you 3 times yesterday, as opposed to 6. Where another person's 4 could mean that their best friend didn't sit next to them at lunch." (C38)

Our participants also indicated that they tend to depend on memory and intuition to interpret and understand individual patients. As C33 said, *"I think you probably have a better memory than you think for the people you're interacting with regularly and using that intuition to guide you."* To support their memory, clinicians may take and review progress notes (e.g., C38 said, *"I would also look back between sessions [notes] if I couldn't think of it [relevant patient's information] in the moment."*) or review various text-based patient data in disjointed systems before or after a session, such as patients' intake assessment in a clinical research database and session notes in patients' electronic health record (EHR). Participant C35 mentioned that *"documents like a skill worksheet don't go into the medical record, but I tend to hold onto them across treatment so that we can refer back to them."*

Participants expressed that they not only need to obtain but also maintain accurate understanding of patients throughout treatment,

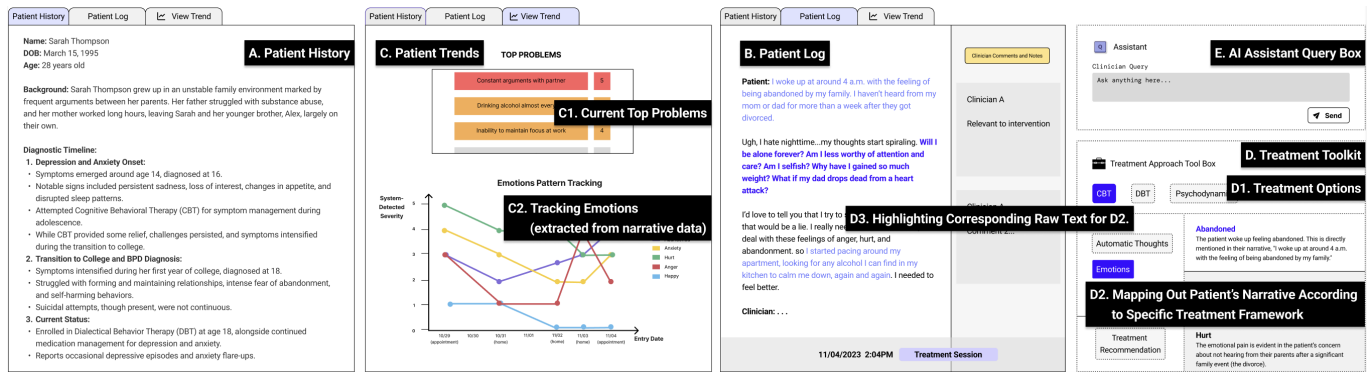


Figure 1: Interactive Figma prototype used in design provocation sessions. A) Patient History shows patient’s profile, diagnoses, and treatments. **B) Patient Log** shows a raw patient narrative. **C) Patient Trends** shows visualized tracking of key information extracted from patient narratives by a language model (LM). **D) Treatment Toolkit** provides LM-generated clinical insights based on different psychotherapy frameworks. **D2 and D3**, for example, show an LM highlighting patients’ thoughts within the text data. Text content shown in the prototype is from a “developed” patient narrative (created by the coauthors) combined with GPT-4 outputs, to simulate using an LM-based tool to derive insights based upon the narratives.

which highlights an opportunity for LM-based tools: to quantify and summarize extensive qualitative patient data to aid clinicians’ monitoring of patients’ conditions and progress throughout psychotherapy. Indeed, in Activity 2, when interacting with the *Patient Trend* feature, which presented a patient’s top problems and a temporal view of the patient’s emotions, clinicians envisioned the utility of monitoring quantified qualitative patient data (i.e., from patient narrative data) over time. As C35 stated, “*You see how individual emotions fluctuate, and how those might relate to other emotions.*”

4.2 Opportunity 2: To provide clinicians a structured tool to support patients’ learning

Our participants described another potential utility of the *Treatment Toolkit* feature (Fig. 1- D) on the prototype : to help patients learn to identify and communicate their automatic thoughts, emotions, and behaviors. Specifically, participants practicing CBT (e.g., C38) highlighted that presenting patients with examples of their maladaptive (or modified) thoughts, emotions, and behavior from their narratives can help patients better understand the rationale for the treatment approach and provide motivation for change. As C38 stated, “*The basis of CBT is to catch your thoughts and behaviors,*” and “*sometimes patients don’t see the patterns and thoughts.*” Similarly, another participant, C39, who employs a psychodynamic approach in psychotherapy, stated that being able to understand and articulate one’s thought process indicates substantial progress in treatment, since “*part of the treatment goal is for one’s narratives to become more accurate and reflective of their experience.*” Thus, our participants envisioned leveraging LM-based tools to better support patients’ learning over the course of treatment. Moreover, C35 and C43 discussed that the mental health treatment process is a collaboration between patients and clinicians, from treatment goal-setting (e.g., what to focus on during treatment) to the treatment approach (e.g., what kind of therapy a patient receives). The quantified and summarized qualitative data was seen to support this process. For

example, C38 described how the simulated LM tool could assist patients with reviewing and reflecting on treatment progress:

“So when you’re reviewing a treatment plan, or if the patient is like ‘I’m not getting better. This isn’t working.’ According to this trend [in the tool], they’re not getting better, but hopefully there will be some corresponding notes to give us what happened between the sessions...It can be helpful when you can track it.” (C38)

4.3 Opportunity 3: To support treatment customization and effectiveness

Finally, we found that LM-based tools could effectively augment psychotherapists’ practice by supporting treatment customization for more targeted and personalized treatments. First, clinicians’ training and treatment orientation shaped their focus and their approach to interpreting patients’ narratives. As C43 said, “*treatment orientation really does change what clinicians attend to and what they emphasize.*” In Activity 1, participants practicing CBT (C34, C35, C43) highlighted different phrases than the clinician practicing psychodynamic (C39) therapy, whereas C38 used a more integrated approach that merged elements of each treatment type (Fig. 2.). Thus, it became apparent that LM tools would need to be customized and consumable to different clinicians, meeting psychotherapists’ varied needs.

Second, C35, C38, and C43 noted that supporting features such as “mapping out patients’ thoughts (Fig. 1-D2)” and “recommending potential treatment interventions (Fig. 1-D)” could be helpful for training new clinicians treatment strategies grounded in a specific therapeutic model. In addition, C35, who not only provided CBT but also taught psychotherapists, noted that processing narrative data could also benefit clinicians learning to adopt alternative treatment approaches. These support tools could enable rapidly applying new treatment frameworks by breaking therapeutic methodologies

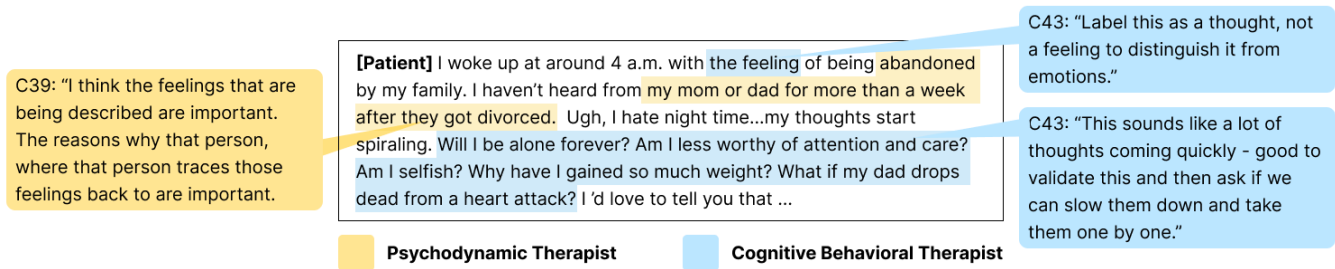


Figure 2: An illustration showing how clinicians with cognitive behavioral (blue) vs. psychodynamic (yellow) approaches attend to different elements of the "developed" patient narrative in Activity 1 and interpret them.

into smaller steps. Regardless of therapists' experience level, well-designed support tools could help augment clinicians' practice and increase clinical efficiency.

Third, LM-based tools could help clinicians tailor one-size-fits-all evidence-based treatments to individual patients. Although clinicians might employ a specific, manualized evidence-based treatment, it is often necessary to tailor the clinicians' treatment approach to best meet the patient's needs (e.g. adjusting treatment priorities or spending longer on a given skill to ensure patient mastery). Through engagement with these paralleled tools that process patient's narratives based on different treatment approach (Fig. 1-D), participants highlighted the need for facilitating flexibility in psychotherapy while allowing clinicians to simultaneously maintain an overarching focus on specific treatment frameworks:

"Flexibility within fidelity is doing things being flexible, but sticking to the spirit of the treatment. I think tools like this could help clinicians work with what patients are giving instead of sticking to what a clinician prepared for the session. If I have two options sort of present to my patients and one approach doesn't work for them, there will be a second skill or intervention approach ready to go." (C35)

As mentioned by C38, such a personalized psychotherapy would help clinicians effectively use the precious and often costly therapeutic time for care.

5 DISCUSSION

Through design provocation, we found that LM-based tools have the potential to augment psychotherapy by providing 1) more accurate measurements and monitoring of patients' subjective experience; 2) an overview of treatment progress to support patients' reflection and learning; and 3) the ability to flexibly adjust treatment approach to best fit patients' needs, thereby facilitating personalized care for patients. In this discussion, we contextualize our results within the current literature, and highlight areas for continued research.

5.1 Measuring and Monitoring Patient-Specific Symptoms, Goals, and Outcomes in Psychotherapy

Our findings in section 4.1 indicate that clinicians combine insights from qualitative patient data, such as patient narratives, with quantitative measures to produce clinically meaningful measurements. These insights help clinicians better understand patients' experiences with their symptoms and formulate more informed treatment plans. Indeed, ongoing research in Natural Language Processing (NLP) and social computing focus on examining patients' linguistic patterns (e.g., those found in patients' social media [37]) to identify symptom severity or state (e.g., manic vs. depressed episodes in bipolar disorder [11]) and to assess treatment response [3]. Our work extends these efforts by demonstrating how LM-based tools can be implemented in psychotherapy for processing relevant treatment-related text, thereby helping clinicians understand patients' unique symptoms and goals to facilitate a more data-driven treatment [2].

Our results affirm existing research on the clinical utility of patient-specific measurements [16, 26, 45, 60], and support patients' learning. Evidently, clinical studies demonstrate that patient-specific outcome measures [52] and data visualization in clinical care [12] might provide patients with validation and encouragement, fostering self-reflection and improving treatment engagement. While future work is required to substantiate LMs' ability to detect and quantify patterns in other forms of treatment-relevant text data (e.g., semi-structured text data from treatment worksheet) produced by patients and others (e.g., parents of young patients) [24]), this study illustrates that LM-based tools could help clinicians discern skills that a patient is likely to practice, evaluate treatment effectiveness, and leverage this information to optimize patient outcomes.

Lastly, our findings motivate us to further explore how LM-based tools can enhance psychotherapy by personalizing treatment. Clinicians indicated that LM-based features that allow extracting and highlighting of clinically-relevant information from patient narratives could help both new and experienced therapists in mastering treatment frameworks and tailoring their approach to better fit patients' needs (Section 4.3). These findings suggest that LM-based tools could serve as novel solutions [15] for disseminating empirically grounded treatment models and balancing their delivery with personalization [5], thereby improving treatment effectiveness.

5.2 Designing Real-World AI Applications to Augment Psychotherapy

Our study employed human-centered design methods to identify design targets for language models to enhance psychotherapy. We included therapists' perspectives in the design process to build clinically meaningful technologies. In this study, language models serve as an all-encompassing underlying structure for prototyping innovative experience, generating output content that can support various design affordances and elicit design needs for the future. Through our analysis, we detailed how LM-powered functionalities can be potentially integrated in a psychotherapist-facing interface. Currently, only a limited number of studies involve psychotherapists in the design process to investigate their perceptions of language models, which may inhibit integrating these technologies into clinical practice [57]. We call for more interdisciplinary human-centered design studies for future LM-based tools. Researchers may consider applying human-centered AI design frameworks such as the task delegability [36] to systematically study how clinicians perceive risk, trust, and utility of delegating different tasks to AI. In our study, for instance, the features that quantified narrative data for tracking and monitoring appeared to pique clinicians' interest (Section 4.1.). The opportunities (Section 4) we identified can serve as a starting point to inspire future systematic need-finding studies and the prototyping of LM-powered tools for psychotherapy.

Last but not least, we acknowledge the limitations and risks associated with AI technologies such as large language models. Our prototype assumed patients' comfort with recording and processing their narrative data using LMs, which might cause privacy and feasibility concerns in practice. Also, LM-caused deficiencies such as misinterpretation of patient data through hallucinations – or factual inaccuracies – may lower clinicians' trust in AI tools. When developing and deploying AI tools for psychotherapy, privacy and safety should be prioritized in evaluation, then benefit and efficacy to treatment outcome [59]. Also, this study was preliminary and exploratory with a small number of clinicians (N=6) being interviewed. In line with the qualitative tradition, the findings should not be considered broadly generalizable. In the design provocation study, we tested a single, limited use case, specifically a Figma prototype where the text content shown to participants was pre-defined.

6 CONCLUSION

Through design provocation sessions with six mental health clinicians, we uncovered three opportunities to augment psychotherapy using language models: quantifying and summarizing patient narratives for easier retrieval and monitoring, providing a structured support tool for patient learning, and facilitating treatment personalization. These results highlight exciting future collaborations between AI, HCI, and clinical researchers to design tools that are not only advanced technologically, but also aligned with psychotherapeutic practice.

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